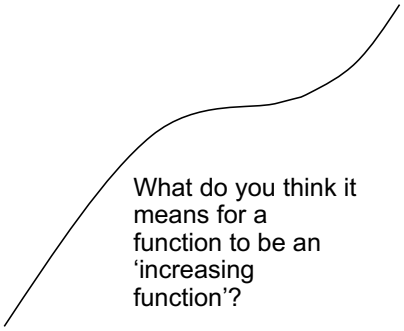
1.34 Differentiation: Stationary Points

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Increasing and Decreasing Functions

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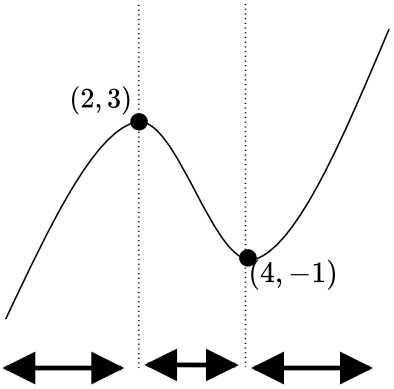


What do you think it means for a function to be an 'increasing function'?

✎ An increasing function is one whose gradient is always at least 0.
$f'(x) \geq 0$ for all x.

It would be '**strictly increasing**' if $f(x) > 0$ for all x, i.e. is not allowed to go horizontal.

A function can also be increasing and decreasing in certain intervals.



We could also write "$f(x)$ is decreasing in the interval $[2, 4]$"
$[a,b]$ represents all the real numbers between a and b inclusive, i.e:

$$[a, b] = \{x : a \leq x \leq b\}$$

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Examples

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Show that the function $f(x) = x^3 + 6x^2 + 21x + 2$ is increasing for all real values of x.

Find the interval on which the function $f(x) = x^3 + 3x^2 - 9x$ is decreasing.

Tip: To show a quadratic is always positive, complete the square, then indicate the squared term is always at least 0.

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Test Your Understanding

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Show that the function $f(x) = x^3 + 16x - 2$ is increasing for all real values of x.

Find the interval on which the function $f(x) = x^3 + 6x^2 - 135x$ is decreasing.

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3 Show that the function $\mathrm{f}(x) = 4 - x(2x^2 + 3)$ is decreasing for all $x \in \mathbb{R}$.

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4 a Given that the function $f(x) = x^2 + px$ is increasing on the interval $[-1, 1]$, find one possible value for p.
b State with justification whether this is the only possible value for p.

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Second Order Derivatives

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When you differentiate once, the expression you get is known as the **first derivative**. Unsurprisingly, when we differentiate a second time, the resulting expression is known as the **second derivative**. And so on…

Original Function	First Derivative	Second Derivative

$y = x^4$

Leibniz's: $\frac{dy}{dx} = 4x^3 \rightarrow \frac{d^2y}{dx^2} = 12x^2$

$y\prime = 4x^3 \rightarrow y\prime\prime = 12x^2$

Newton's: $\dot{y} = 4x^3 \rightarrow \ddot{y} = 12x^2$

Lagrange's: $f(x) = x^4 \rightarrow f\prime(x) = 4x^3 \rightarrow f\prime\prime(x) = 12x^2$

You can similarly have the third derivative ($\frac{d^3y}{dx^3}$), although this is no longer in the A Level syllabus. We'll see why might use the second derivative soon…

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Just for your interest…

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How does the notation $\frac{d^2y}{dx^2}$ work?
Why are the squareds where they are?

Suppose that $y = x^3 + 1$.
Then when we write $\frac{dy}{dx}$, we're effectively doing $\frac{d(x^3+1)}{dx}$ (by substitution), although this would typically be written:
$\frac{d}{dx}\left(x^3 + 1\right)$

The $\frac{d}{dx}(\dots)$ notation is quite handy, because it behaves as a function and allows us to write the original expression and the derivative within a single equation:
$\frac{d}{dx}\left(x^3 + 1\right) = 3x^2$

Therefore, if we wanted to differentiate y twice, we'd do:
$\frac{d}{dx}\left(\frac{d}{dx}(y)\right)$
$= \frac{d^2}{dx^2}(y)$
$= \frac{d^2y}{dx^2}$

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Examples

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If $y = 3x^5 + \frac{4}{x^2}$, find $\frac{d^2y}{dx^2}$.

If $f(x) = 3\sqrt{x} + \frac{1}{2\sqrt{x}}$, find $f''(x)$.

1.34 Differentiation: Stationary Points

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Test Your Understanding

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If $y = 5x^3 - \frac{x}{3\sqrt{x}}$, find $\frac{d^2y}{dx^2}$.

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3 Given that $y = (2x - 3)^3$, find the value of x when $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$.

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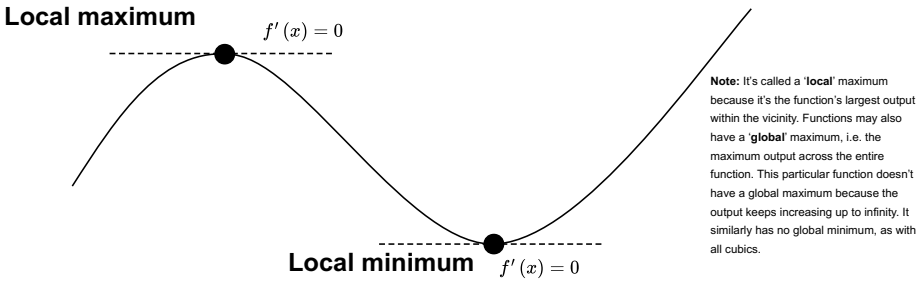
4\quad $\mathrm{f}(x) = px^3 - 3px^2 + x^2 - 4$. When $x = 2$, $\mathrm{f}''(x) = -1$. Find the value of p.

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Stationary/Turning Points

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A stationary point is where the gradient is 0, i.e. $f'(x) = 0$.



Note: It's called a '**local**' maximum because it's the function's largest output within the vicinity. Functions may also have a '**global**' maximum, i.e. the maximum output across the entire function. This particular function doesn't have a global maximum because the output keeps increasing up to infinity. It similarly has no global minimum, as with all cubics.

Find the coordinates of the turning points of $y = x^3 + 6x^2 - 135x$

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More Examples

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Find the least value of $f(x) = x^2 - 4x + 9$

Find the turning point of $y = \sqrt{x} - x$

Method 1: Differentiation

Method 2: Completing the square

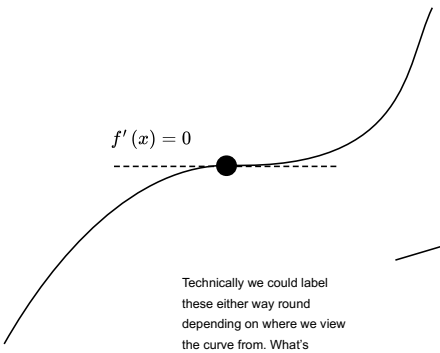
Note: Method 2 is only applicable for quadratic functions. For others, differentiation must be used.

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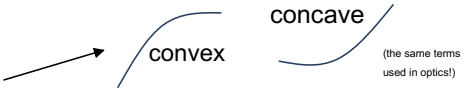
Points of Inflection

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There's a third type of stationary point (that we've encountered previously):



A point of inflection is where the curve changes from convex concave (or vice versa).

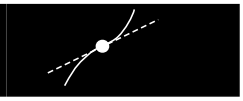


(the same terms used in optics!)

Technically we could label these either way round depending on where we view the curve from. What's important is that the **concavity changes**.

i.e. the line curves in one direction before the point of inflection, then curves in the other direction after.

Side Note: Not all points of inflection are stationary points, as can be seen in the example on the right.
A point of inflection which **is** a stationary point is known as a *saddle point*.



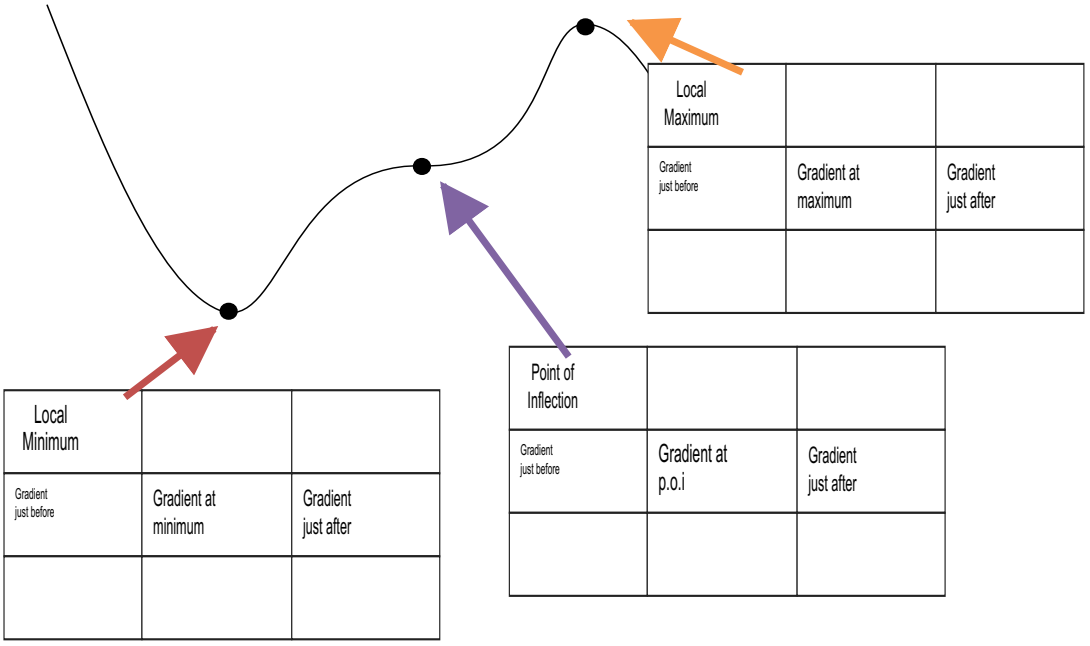
1.34 Differentiation: Stationary Points

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How do we tell what type of stationary point?

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Method 1: Look at gradient just before and just after point.



Local Maximum		
Gradient just before	Gradient at maximum	Gradient just after

Point of Inflection		
Gradient just before	Gradient at p.o.i	Gradient just after

Local Minimum		
Gradient just before	Gradient at minimum	Gradient just after

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How do we tell what type of stationary point?

52

Method 1: Look at gradient just before and just after point.

Find the stationary point on the curve with equation $y = x^4 - 32x$, and determine whether it is a local maximum, a local minimum or a point of inflection.

Strategy: Find the gradient for values just before and after $x = 2$. Let's try $x = 1.9$ and $x = 2.1$.

	x=1.9	x=2	x=2.1
Gradient			
Shape			

1.34 Differentiation: Stationary Points

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Method 2: Using the second derivative

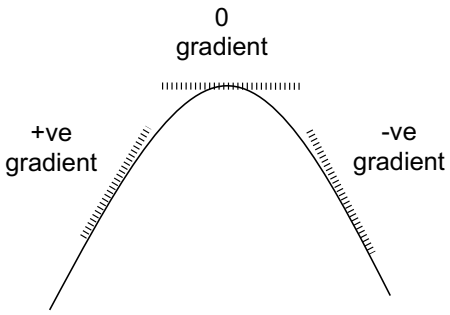
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The method of substituting values of x just before and after is a bit cumbersome.
It also has the potential for problems: what if two different types of stationary points are really close together?

Recall the gradient gives a measure of the **rate of change** of y, i.e. how much the y value changes as x changes.

Thus by differentiating the gradient function, the **second derivative tells us the rate at what the gradient is changing**.

Thus if the second derivative is positive, the gradient is increasing.
If the second derivative is negative, the gradient is decreasing.



At a maximum point, we can see that as x increases, the gradient is decreasing from a positive value to a negative value.

$$\therefore \frac{d^2y}{dx^2} < 0$$

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Method 2: Using the second derivative

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✎ At a stationary point $x = a$:
- If $f''(a) > 0$ the point is a local minimum.
- If $f''(a) < 0$ the point is a local maximum.
- If $f''(a) = 0$ it could be any type of point, so resort to Method 1.

The stationary point of $y = x^4 - 32x$ is $(2, -48)$. Use the second derivative to classify this stationary point.

I will eventually do a 'Just for your Interest…' thingy on why we can't classify the point when $f''(x) = 0$, and how we could use the **third derivative**!

Find $\frac{d^2y}{dx^2}$ for the x at the stationary point.

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Test Your Understanding

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Edexcel C2 May 2013 Q9

The curve with equation $y = x^2 - 32\sqrt{x} + 20$, $x > 0$, has a stationary point P. Use calculus

(a) to find the coordinates of P, (6)

(b) to determine the nature of the stationary point P. (3)

(a)

(b)

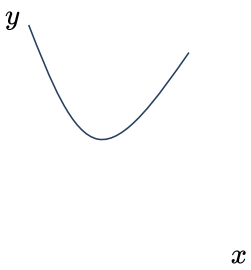
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Sketching Graphs

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All the way back in Chapter 4, we used features such as intercepts with the axes, and behaviour when $x \to \infty$ and $x \to -\infty$ in order to sketch graphs. Now we can also find stationary/turning points!

[Textbook] By first finding the stationary points, sketch the graph of $y = \frac{1}{x} + 27x^3$



? As $x \to \infty$, $x \to -\infty$



1.34 Differentiation: Stationary Points

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5 By considering the gradient on either side of the stationary point on the curve $y = x^3 - 3x^2 + 3x$, show that this point is a point of inflection. Sketch the curve $y = x^3 - 3x^2 + 3x$.

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6 Find the maximum value and hence the range of values for the function $\mathrm{f}(x) = 27 - 2x^4$.

1.34 Differentiation: Stationary Points

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7\quad $\mathrm{f}(x) = x^4 + 3x^3 - 5x^2 - 3x + 1$.
a Find the coordinates of the stationary points of $\mathrm{f}(x)$, and determine the nature of each.
b Sketch the graph of $y = \mathrm{f}(x)$.

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Extension

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1 [MAT 2014 1C] The cubic $y = kx^3 - (k+1)x^2 + (2-k)x - k$ has a turning point, that is a minimum, when $x = 1$ precisely for
A) $k > 0$
B) $0 < k < 1$
C) $k > \frac{1}{2}$
D) $k < 3$
E) all values of k
F)

2 *[MAT 2004 1B]* The smallest value of the function:
$f(x) = 2x^3 - 9x^2 + 12x + 3$
In the range $0 \le x \le 2$ is what?

3 *[MAT 2001 1E]* The maximum gradient of the curve $y = x^4 - 4x^3 + 4x^2 + 2$ in the range $0 \le x \le 2\frac{1}{5}$ occurs when:
A) $x = 0$
B) $x = 1 - \frac{1}{\sqrt{3}}$
C) $x = 1 + \frac{1}{\sqrt{3}}$
D) $x = 2\frac{1}{5}$
E)

Hint: When two curves touch, their y values must match, but what else must also match?

[STEP I 2007 Q8] A curve is $y = ax^3 - 6ax^2 + (12a+12)x - (8a+16)$

4 where a is a real number. Show that this curve touches the curve with equation $y = x^3$ at $(2, 8)$. Determine the coordinates of any other point of intersection of the two curves.
(i) Sketch on the same axes these two curves when $a = 2$.
(ii) … when $a = 1$ (iii) when $a = -2$

1.34 Differentiation: Stationary Points

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Solutions to Extension Questions 61

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Solutions to Extension Questions 62

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Solutions to Extension Questions

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